

margin of less than 1%, all through non-contact measurement. For anyone involved in EV design, this translates to efficient battery monitoring with a small, reliable sensor that minimises power consumption.

And if you are working with wireless EV charging, TMR sensors have more to offer. Wireless systems require accurate data from the magnetic fields between the transmitting and receiving coils to function properly. This is where TMR sensors excel. Placed beneath the receiving coil, they monitor the magnetic field in real time, detecting performance, alignment, and potential obstructions. The result? A system capable of identifying misalignment or metal interference, which can impact charging efficiency.

Traditional wireless charging often uses complex hardware for coil alignment, adding cost and setup challenges. With TMR sensor arrays, misalignment detection becomes straightforward and cost-efficient. When placed on the same plane as the receiving coils, they measure the magnetic field from the transmitting coils, ensuring consistent charging and power transfer—even when alignment is imperfect.

Why choose TMR sensors for security systems?

Magnetic sensing plays a crucial role in today's security systems, especially when it comes to detecting unauthorised entry through doors and windows.

Here is how it works: A TMR sensor typically consists of a magnet and a sensor, placed close to each other, often on a door or window frame. When the door or window is closed, the sensor keeps reading a set magnetic field. Now, picture this: when an intruder opens the door or window, the magnet moves away, which reduces the magnetic field detected by the sensor. That shift

Key players in the TMR sensors market

The TMR sensors market is rapidly expanding, with several key companies leading the way in innovation and market growth. These players provide high-performance solutions across automotive, industrial automation, and consumer electronics industries. The alpha-sorted list of key players includes:

- Allegro Microsystems (US)
- Coto Technology (US)
- Crocus Technology (US)
- Infineon Technologies (Germany)
- MultiDimension Technology (Taiwan)
- NVE Corporation (US)
- Sensitec GmbH (Germany)
- Sinomags (China)
- TDK Micronas (Japan)

These companies are at the forefront, driving demand with their innovative TMR sensors, which cater to applications like electric vehicles, robotics, and IoT, all while helping shape the future of precision sensing technology.

triggers the alarm, letting you know there has been an entry.

Now, for all the manufacturers out there: switching from traditional reed switches to TMR sensors is a breeze. They mount on circuit boards just like reed switches and use similar wiring. So, you do not need to reinvent the wheel; you can upgrade to the more reliable TMR sensor without disrupting your current designs.

One of the great things about TMR sensors is their high sensitivity. Even if the magnet is not perfectly aligned, the sensor will still work accurately. This makes installations much easier, reduces errors, and improves overall reliability. Have you ever had trouble with misalignment during installation? TMR sensors can help avoid those issues, saving time and boosting customer satisfaction.

TMR sensors are also a great fit for battery-powered wireless security systems. Since they consume very little power, they can run for long periods, reducing the need for constant maintenance. TMR sensors offer a longer, more reliable option than reed switches, which have limited lifespan and durability.

How TMR sensors are powering automation

TMR sensors are changing the game in automation. Think about it—whether it is a robot arm in a factory or the power steering in your car, sensors are the quiet heroes behind the scenes, providing the data needed for precise positioning, power management, and overall functionality. In factories, robotic arms use position sensors and motor drivers to pick, place, and adjust components exactly where they need to go. Pretty impressive, right?

But wait, robots are not just in factories—they are appearing everywhere, from warehouses to hospitals, and even at home! And you know what makes them move so smoothly? Motor controllers and angle sensors help them perform tasks with pinpoint accuracy. Service robots, for instance, use sensors to navigate rooms, interact with people, and understand their surroundings. Ever seen a robot adapt to a room layout? That is all thanks to TMR sensors.

These sensors are game-changers when it comes to motor control. They help measure angles, speed, and positioning with remarkable